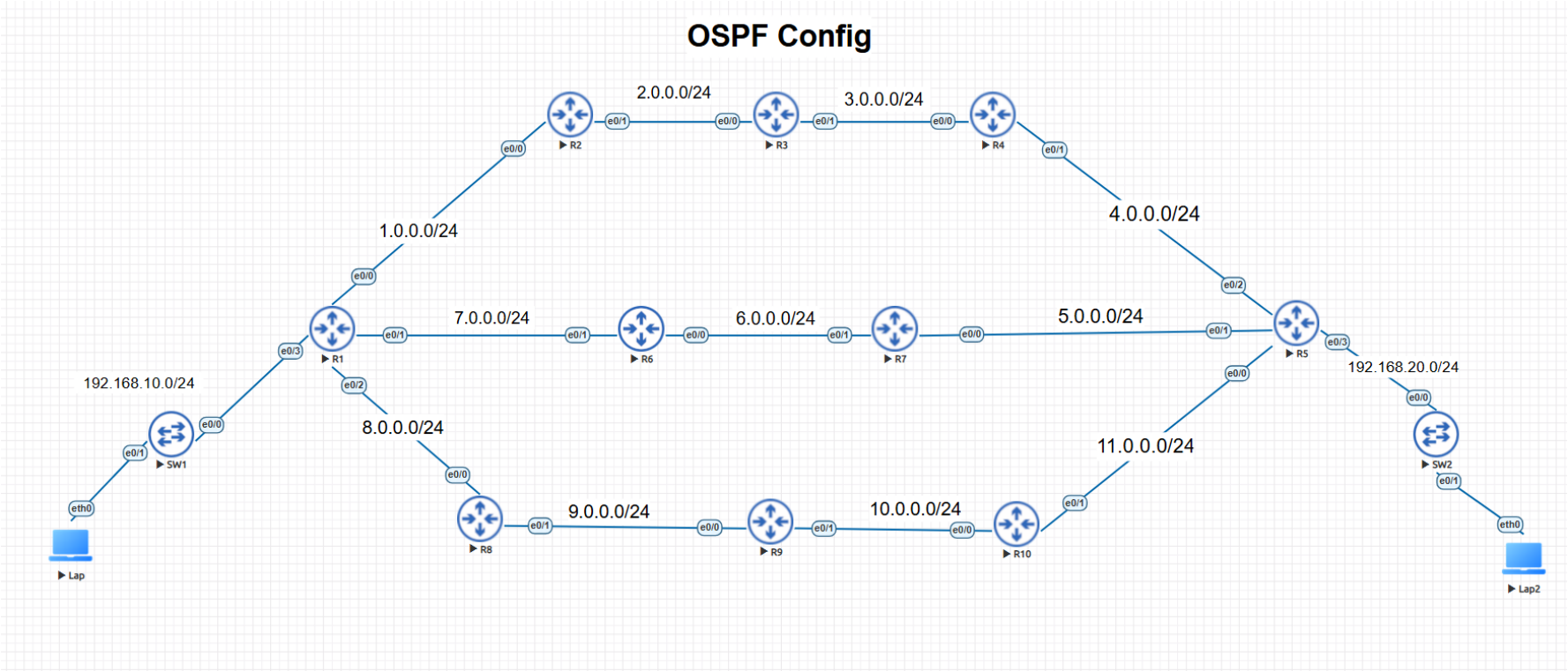
OSPF single-area network lab

Repository documentation | Network design, configuration evidence and end-to-end validation

Ten routers. Three transit branches. One captured route.

The lab connects 192.168.10.0/24 to 192.168.20.0/24 across a ten-router topology. R1 is configured for OSPF process 100 in area 0. The supplied tests show five ping replies and a trace through R1, R6, R7 and R5.



Captured path

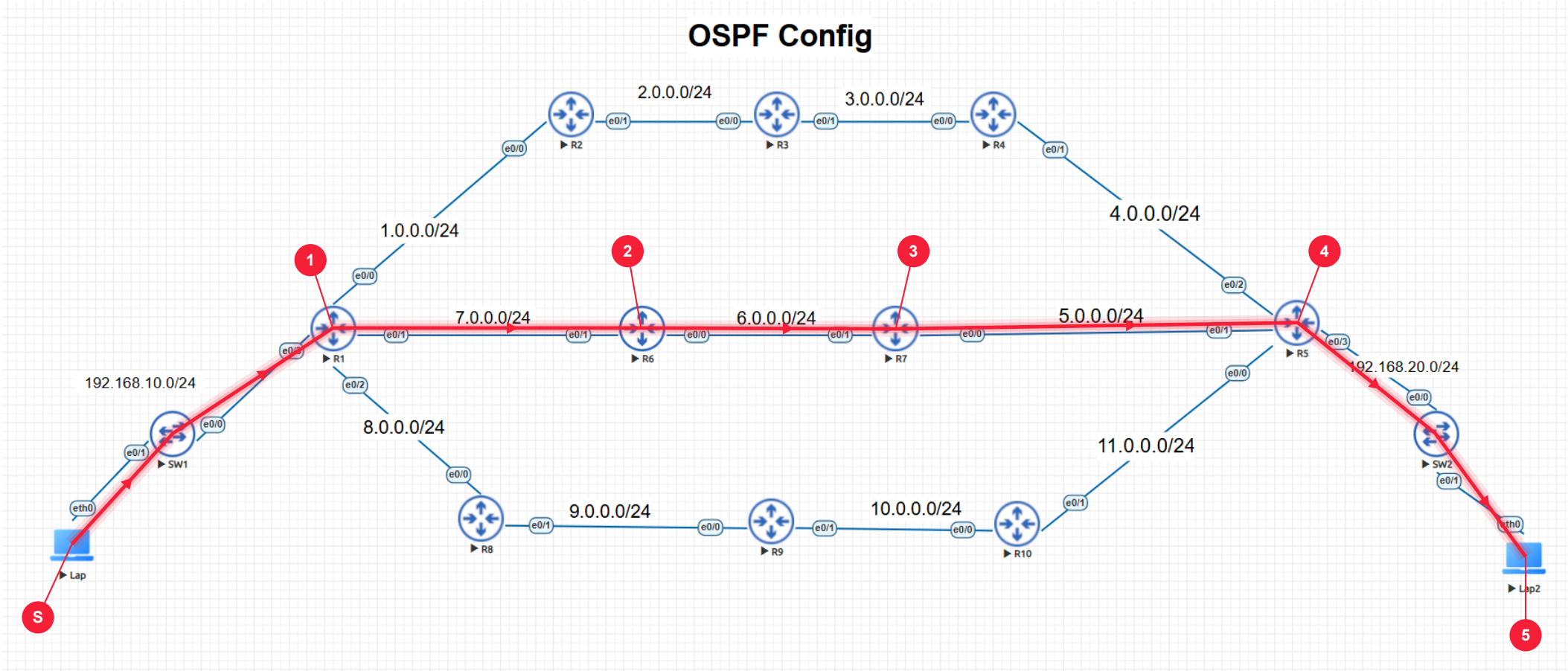
Lap > SW1 > R1 > R6 > R7 > R5 > SW2 > Lap2

Source material

Seven screenshots; configuration and console evidence retained.

Observed path across the original topology

Bright red arrows follow the captured trace. Switches forward the LAN frames without adding IP hops.



S Lap: 192.168.10.1 > 1 R1: 192.168.10.254 > 2 R6: 7.0.0.2
3 R7: 6.0.0.2 > 4 R5: 5.0.0.2 > 5 Lap2: 192.168.20.1

The annotation identifies the path observed by these probes. It does not show the return path or prove that every flow uses the same branch.

Subnet and link inventory

Topology labels show eleven transit /24 networks and two endpoint /24 LANs. All use mask 255.255.255.0.

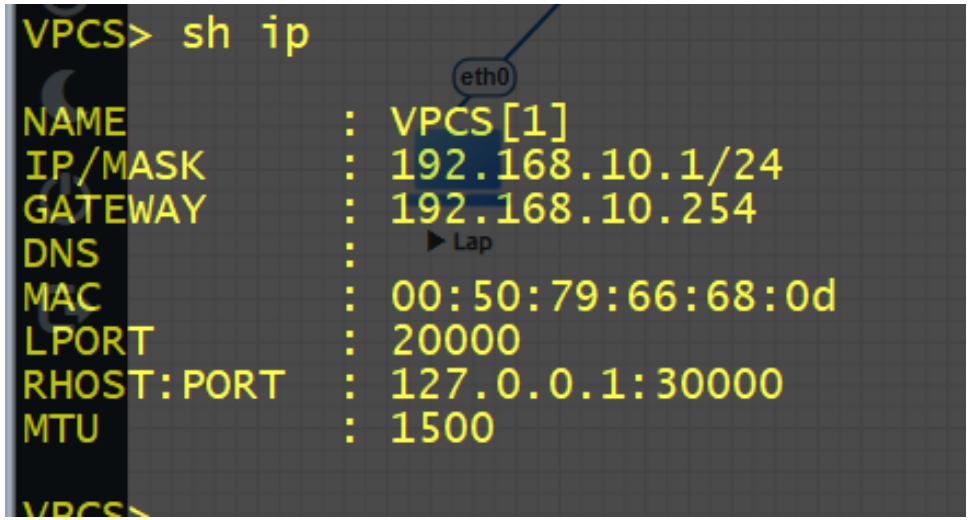
Segment	Subnet	Link / attached devices	Evidence
Source LAN	192.168.10.0/24	Lap - SW1 - R1	Lap .1; R1 .254
Upper 1	1.0.0.0/24	R1 e0/0 - R2 e0/0	R1 .1 captured
Upper 2	2.0.0.0/24	R2 e0/1 - R3 e0/0	Topology label
Upper 3	3.0.0.0/24	R3 e0/1 - R4 e0/0	Topology label
Upper 4	4.0.0.0/24	R4 e0/1 - R5 e0/2	Topology label
Middle 1	7.0.0.0/24	R1 e0/1 - R6 e0/1	R1 .1; trace R6 .2
Middle 2	6.0.0.0/24	R6 e0/0 - R7 e0/1	Trace R7 .2
Middle 3	5.0.0.0/24	R7 e0/0 - R5 e0/1	Trace R5 .2
Lower 1	8.0.0.0/24	R1 e0/2 - R8 e0/0	R1 .1 captured
Lower 2	9.0.0.0/24	R8 e0/1 - R9 e0/0	Topology label
Lower 3	10.0.0.0/24	R9 e0/1 - R10 e0/0	Topology label
Lower 4	11.0.0.0/24	R10 e0/1 - R5 e0/0	Topology label
Destination LAN	192.168.20.0/24	R5 - SW2 - Lap2	Lap2 .1; gateway .254

e0/x abbreviates Ethernet0/x. R1 e0/3 serves the source LAN; R5 e0/3 serves the destination LAN. Unshown interface addresses are not inferred from subnet labels.

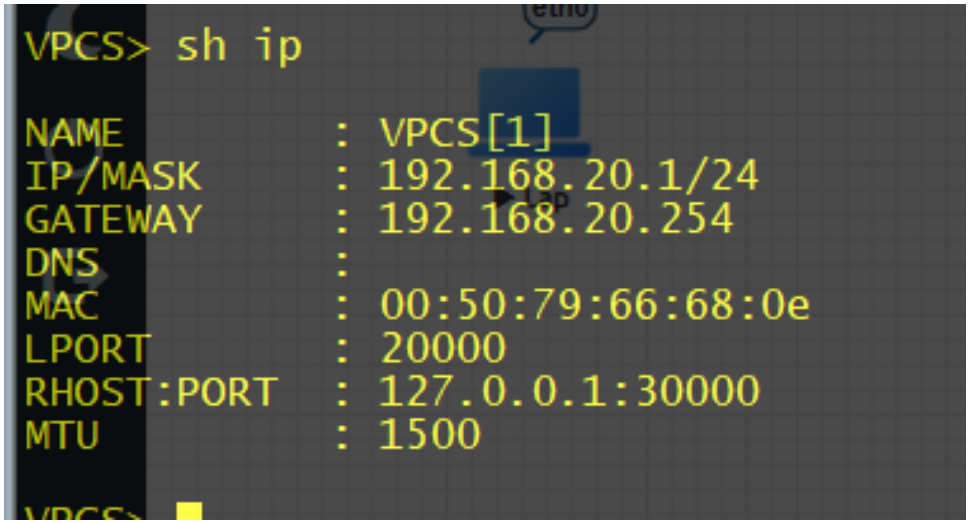
Host addressing and default gateways

The VPCS show-IP captures confirm both endpoint configurations.

Lap - source host



Lap2 - destination host

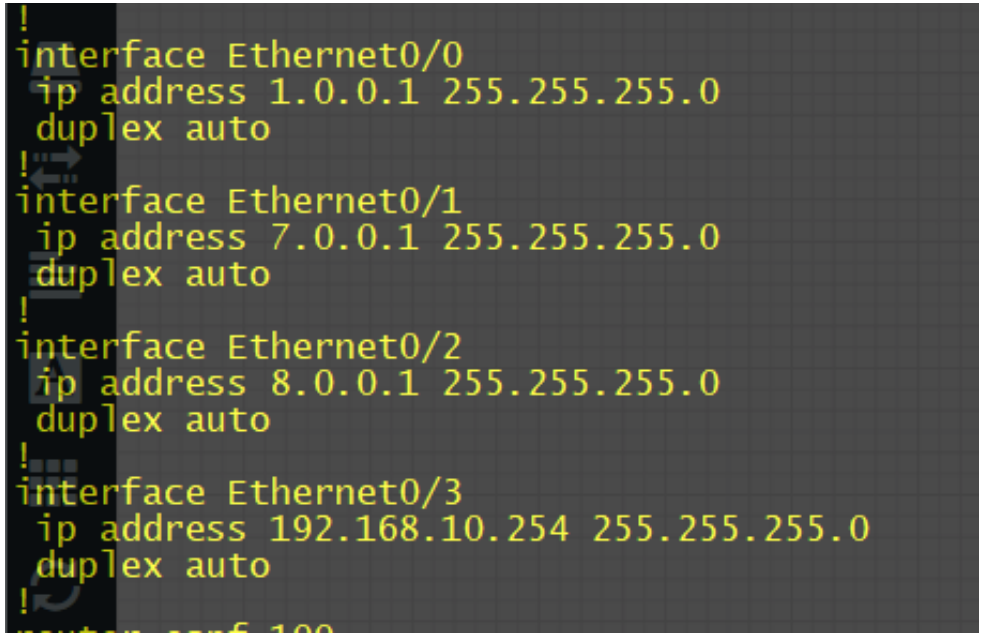


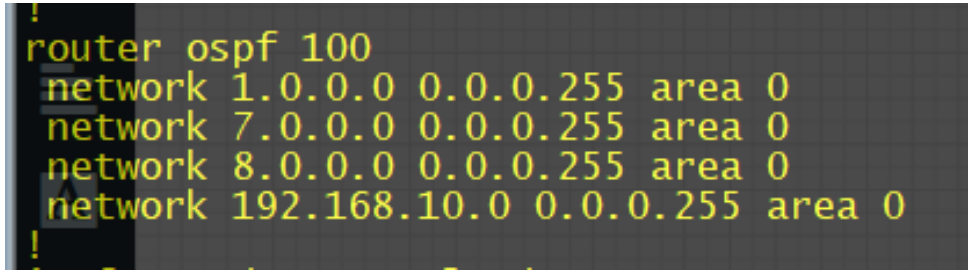
Host	IPv4 address	Mask	Default gateway
Lap	192.168.10.1	255.255.255.0	192.168.10.254
Lap2	192.168.20.1	255.255.255.0	192.168.20.254

For a remote subnet, each host sends its traffic to the configured LAN gateway. SW1 and SW2 provide local switching between the hosts and their edge routers. The screenshots do not include switch configuration or VLAN details.

R1: captured interface and OSPF settings

The interface addresses match R1 in the topology. The screenshots show OSPF process 100 and four area 0 statements.





Reproduction snippet for R1

```
enable
configure terminal
interface Ethernet0/0
 ip address 1.0.0.1 255.255.255.0
 no shutdown
interface Ethernet0/1
 ip address 7.0.0.1 255.255.255.0
 no shutdown
interface Ethernet0/2
 ip address 8.0.0.1 255.255.255.0
 no shutdown
interface Ethernet0/3
 ip address 192.168.10.254 255.255.255.0
 no shutdown
exit
router ospf 100
 network 1.0.0.0 0.0.0.255 area 0
 network 7.0.0.0 0.0.0.255 area 0
 network 8.0.0.0 0.0.0.255 area 0
 network 192.168.10.0 0.0.0.255 area 0
end
copy running-config startup-config
```

IP addresses and OSPF network statements are captured settings. Mode changes, no shutdown and saving are added reproduction steps. The original extract also shows duplex auto.

Extend area 0 across the topology

Suggested configuration only: peer running configurations and OSPF neighbor tables were not supplied.

Middle branch and destination edge

```
! R6 - apply separately
router ospf 100
 network 7.0.0.0 0.0.0.255 area 0
 network 6.0.0.0 0.0.0.255 area 0
! R7 - apply separately
router ospf 100
 network 6.0.0.0 0.0.0.255 area 0
 network 5.0.0.0 0.0.0.255 area 0
! R5 - apply separately
router ospf 100
 network 4.0.0.0 0.0.0.255 area 0
 network 5.0.0.0 0.0.0.255 area 0
 network 11.0.0.0 0.0.0.255 area 0
 network 192.168.20.0 0.0.0.255 area 0
```

Upper and lower branch networks

Router	Networks to match in area 0
R2	1.0.0.0/24; 2.0.0.0/24
R3	2.0.0.0/24; 3.0.0.0/24
R4	3.0.0.0/24; 4.0.0.0/24
R8	8.0.0.0/24; 9.0.0.0/24
R9	9.0.0.0/24; 10.0.0.0/24
R10	10.0.0.0/24; 11.0.0.0/24

Before applying the peer snippets

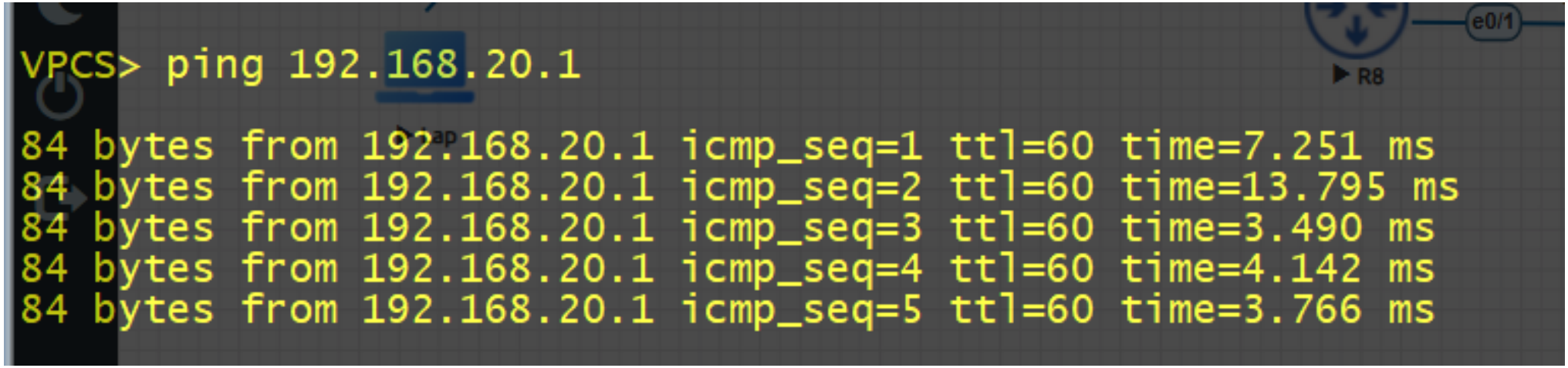
Assign unique addresses to every connected interface, enable the links and verify each peer subnet. Use 192.168.20.254/24 on R5 e0/3 to match Lap2's captured gateway.

For each listed subnet, use its network address with wildcard 0.0.0.255 in a network statement under router ospf 100. The wildcard selects matching local interfaces for area 0. [1]

Enter each block separately in global configuration mode on its named router, then exit and save. OSPF process IDs are locally significant; using 100 throughout is a convention. Verify interface addressing and adjacency settings before expecting learned routes. [1]

Five replies from the destination host

Original VPCS output: ping 192.168.20.1. The screenshot contains five consecutive successful reply lines.

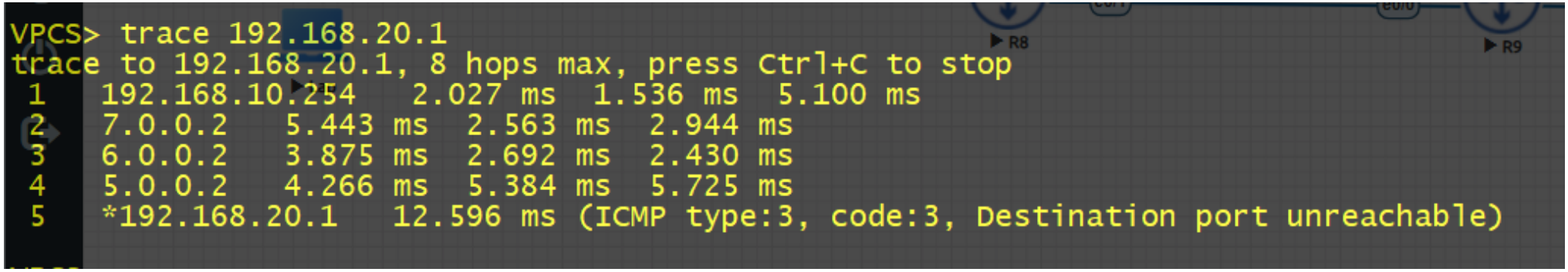


Sequence	Reply source	Displayed size	TTL	Round-trip time
1	192.168.20.1	84 bytes	60	7.251 ms
2	192.168.20.1	84 bytes	60	13.795 ms
3	192.168.20.1	84 bytes	60	3.490 ms
4	192.168.20.1	84 bytes	60	4.142 ms
5	192.168.20.1	84 bytes	60	3.766 ms

All five displayed probes receive replies. Their RTTs range from 3.490 to 13.795 ms, with a calculated mean of 6.489 ms. The screenshot shows no final loss summary, so these five lines should not be presented as a longer-term availability measurement.

The middle branch, confirmed by trace

Original VPCS output: trace 192.168.20.1, with a maximum of 8 hops.



Hop	Device / responder	Probe 1	Probe 2	Probe 3
1	R1 - 192.168.10.254	2.027 ms	1.536 ms	5.100 ms
2	R6 - 7.0.0.2	5.443 ms	2.563 ms	2.944 ms
3	R7 - 6.0.0.2	3.875 ms	2.692 ms	1.430 ms
4	R5 - 5.0.0.2	4.266 ms	5.384 ms	5.725 ms
5	Lap2 - 192.168.20.1	12.596 ms	Not shown	Not shown

Why the last line says port unreachable. The reply comes from the destination address, 192.168.20.1. ICMP type 3, code 3 means destination port unreachable. In this UDP trace, that response indicates the probe reached the destination, where no application accepted the probed port; it does not contradict the successful ping. [2]

Route selection and verification

OSPF uses path cost. The drawing alone cannot establish the actual configured costs or installed routes.

Branch	Router path from R1 to R5	Transit links
Upper	R1 > R2 > R3 > R4 > R5	4
Middle - observed	R1 > R6 > R7 > R5	3
Lower	R1 > R8 > R9 > R10 > R5	4

If transit interface costs are equal, the middle branch has the lowest total transit cost. This is consistent with the trace. OSPF selects the lowest accumulated cost, rather than simply counting routers; interface costs or policy could change the result. The common destination LAN cost does not change this branch comparison. [1]

Capture these outputs to establish the control-plane state

```
show ip interface brief
show ip ospf neighbor
show ip ospf interface brief
show ip route ospf
show ip route 192.168.20.1
```

On each router, check interfaces, neighbor adjacencies, area membership and OSPF routes. Inspect interface costs with show ip ospf interface. A trace proves the forwarding path for its probes; routing-table evidence is needed to confirm that OSPF installed it.

For failover validation, capture a baseline, disable one middle-branch link in the lab, wait for convergence, then repeat ping and trace. Restore the link and record recovery. Failover and convergence time were not tested in the supplied evidence.

Results, scope and repository use

Evidence-based conclusions, with the original screenshots retained throughout this guide.

What the screenshots establish

The endpoint LAN addresses and gateways are configured. R1 has OSPF process 100 with four area 0 matches. Five displayed ping probes receive replies. Trace reaches Lap2 through R1, R6, R7 and R5.

What remains to verify

Complete peer addressing, OSPF adjacencies, installed route sources, interface costs, reverse-path selection and failover behavior are not shown. No simulator project or live console session was available for additional tests.

Suggested repository placement

```
docs/OSPF_Network_Documentation.pdf
images/OSPF_Annotated_Traceroute.png
```

Link the PDF and annotated image from the repository README. Keep the simulator project and verified device configurations alongside the documentation when available. The paths above are suggested repository locations, not files published by this session.

References and provenance

[1] Cisco OSPF Configuration Guide - OSPF interface matching, process configuration and cost-based route selection.
[2] RFC 792: Internet Control Message Protocol - Destination Unreachable, type 3 / code 3.

Primary evidence: seven user-supplied screenshots. Topology: pages 1-2; host settings: page 4; R1 configuration: page 5; ping: page 7; trace: page 8. Red paths and hop badges are documentation overlays on the original topology. Suggested configuration is explicitly labeled.